EIGRP Metrics:

EIGRP uses a composite metric based on five parameters:

Bandwidth: The minimum bandwidth of the route.

Delay: The cumulative delay of the route.

Load: The load on the route (optional).

Reliability: The reliability of the route (optional).

MTU: The Maximum Transmission Unit (not used in the metric calculation).

EIGRP Metrics:

The default EIGRP metric formula primarily uses bandwidth and delay:

Metric=((10^7/Minimum Bandwidth)+Cumulative Delay)×256

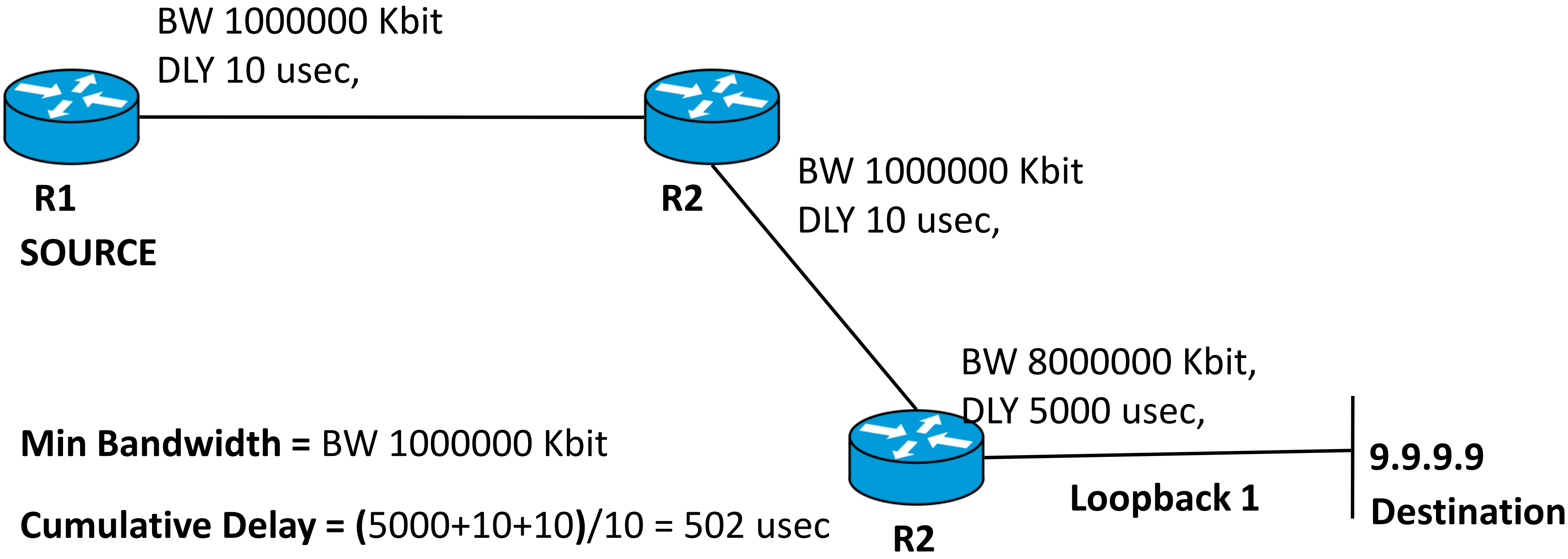
Minimum Bandwidth: the least bandwidth of the recieving end ib between source and destination

Cumulative Delay: (Sum of all delay on receiving end)/10

Command used to check the Delay and Bandwidth of interface:

Router#show int g0/0/0 | Section DLY

EIGRP ROUTE METRIC:



Min Bandwidth = BW 1000000 Kbit

Cumulative Delay = (5000+10+10**)**/10 = 502 usec

Metric=((10^7/Minimum Bandwidth)+Cumulative Delay)×256

Metric=((10^7/1000000)+502)×256= (10+502)×256=512*256=131072

https://www.youtube.com/@techwithravish